Schema Graph — E-Commerce

SQL Clear Run

```
select * from users;
```

11463 rows ×

USER_ID	FN	ACTIVE	CLUB_MEMBER_STATUS	FASHION_NEWS_FREQUENCY	AGE
0	true	true	ACTIVE	Regularly	49
1	false	false	ACTIVE	None	28
2	true	true	ACTIVE	Regularly	26
3	true	true	ACTIVE	Regularly	20
4	true	true	ACTIVE	Regularly	40
5	false	false	ACTIVE	None	57
6	true	true	ACTIVE	Regularly	20
7	true	true	ACTIVE	Regularly	40
8	false	false	ACTIVE	None	null
9	true	true	ACTIVE	Regularly	19
10	false	false	ACTIVE	None	40
11	false	false	ACTIVE	None	24
12	true	true	PRE-CREATE	Regularly	65
13	false	false	ACTIVE	None	70
14	false	false	ACTIVE	None	56
15	false	false	PRE-CREATE	None	null
16	false	false	ACTIVE	None	52
17	true	true	ACTIVE	Regularly	23
18	false	false	ACTIVE	None	26
19	true	false	ACTIVE	Regularly	35

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PQL Clear Explain Evaluate Run

```
PREDICT COUNT(orders.*, 0, 30, days) > 0
FOR EACH users.user_id
```

5000 rows ×

ENTITY	ANCHOR_TIMESTAMP	TARGET_PRED	FALSE_PROB	TRUE_PROB
0	2020-09-19T00:00:00	false	0.5869964361	0.4130035639
1	2020-09-19T00:00:00	false	0.9447912574	0.055208683
2	2020-09-19T00:00:00	false	0.9931228757	0.0068771071
3	2020-09-19T00:00:00	false	0.9912548065	0.0087451823
4	2020-09-19T00:00:00	true	0.4073334336	0.592666626
5	2020-09-19T00:00:00	false	0.8418256044	0.1581743509
6	2020-09-19T00:00:00	false	0.9935117364	0.0064882697
7	2020-09-19T00:00:00	false	0.8860265613	0.1139734611
8	2020-09-19T00:00:00	false	0.9752961397	0.0247038696
9	2020-09-19T00:00:00	false	0.9244152308	0.0755847916
10	2020-09-19T00:00:00	false	0.8479677439	0.1520322412
11	2020-09-19T00:00:00	false	0.7154240012	0.2845759988
12	2020-09-19T00:00:00	true	0.3451052904	0.6548947096
13	2020-09-19T00:00:00	false	0.9439706206	0.0560293831
14	2020-09-19T00:00:00	false	0.8006919622	0.199308008
15	2020-09-19T00:00:00	false	0.9913558364	0.0086441776
16	2020-09-19T00:00:00	true	0.1955940723	0.8044059277
17	2020-09-19T00:00:00	false	0.9739436507	0.0260563046
18	2020-09-19T00:00:00	true	0.1907240152	0.8092759252
19	2020-09-19T00:00:00	true	0.2422056347	0.7577943802

1000 rows (truncated) Prev 1 / 50 Next

EXPLANATION

The model predicts that user 4896 is very likely to place at least one order in the next 30 days, with a high True probability of approximately 89.2%. This confident prediction is well supported by both global cohort patterns and the user's local data.

From the **global cohorts**, several points stand out:

- Users with **1+ past orders** have around 19.4% average probability of ordering again; the model here expects a much higher probability for this user, indicating strong individual signals.
- Orders with prices mostly in the **(28.31 - 30.59]** range correlate to a roughly 22.7% chance of ordering, typical of moderately priced items, suggesting likelihood is not driven solely by luxury or bargain pricing.
- Most recent order dates for users influence probability, with **orders in recent months (0 to 2 months)** showing substantially higher reorder likelihood around 35%.
- The user's cohort for **age (17)** shows a lower baseline reorder chance (about 13.4%), implying age alone would not predict high ordering—but other factors compensate.

Looking at the **local subgraph**, the model identifies several impactful factors:

- The user is **not currently active (active=False)** and has **no fashion news subscriptions (fashion_news_frequency=None)**, both moderately decreasing influence (scores ~0.01–0.05).
- However, the user is a **club member with ACTIVE status**, which carries the strongest positive effect among user attributes (score ~0.116), consistent with the global cohort showing active club members have higher reorder rates (~20.6%).
- The key drivers here are the user's **multiple recent product views dated very close to the prediction timestamp**, with several views scoring very high (e.g., scores from ~0.24 up to 0.69). This recency and frequency of views strongly increase reorder probability because the model links recent interest in items to imminent buying.
- Viewed items mostly belong to categories and features associated with moderate to high reorder likelihood, such as:
  - Item 19, a "Garment Lower body, Dark color, Solid pattern" with exceptionally high importance scores (e.g., graphical_appearance_name=1.0, perceived_color_master_name=0.75, product_type_name=0.46). These features align with product cohorts that have reorder targets in the 18–42% range or higher.
  - Other viewed items also contribute moderately via similar apparel characteristics (e.g., dark colors, solid or all-over patterns in "Divided" or "Ladieswear" groups).

In summary, the user's likelihood to place an order soon is mainly boosted by their active club membership and strong recent browsing activity on appealing product types. Despite being young and currently inactive in other ways, the intense recent engagement signals a high purchase intent. The model's high confidence (True probability ~89.2%) reflects these combined influences.

MODEL PERFORMANCE

60.2%	60.1 %	88.4%
AP	AUPRC	AUROC

PQL Clear Explain Evaluate Run

```
# Suggested: How many views will each item have in the next 90 days?
PREDICT COUNT(views.*, 0, 90, days)
FOR EACH items.item_id
```

5000 rows ×

ENTITY	ANCHOR_TIMESTAMP	TARGET_PRED
0	2020-09-19T00:00:00	0
1	2020-09-19T00:00:00	0
2	2020-09-19T00:00:00	0
3	2020-09-19T00:00:00	0
4	2020-09-19T00:00:00	0
5	2020-09-19T00:00:00	0
6	2020-09-19T00:00:00	0
7	2020-09-19T00:00:00	0
8	2020-09-19T00:00:00	0
9	2020-09-19T00:00:00	0
10	2020-09-19T00:00:00	0
11	2020-09-19T00:00:00	0
12	2020-09-19T00:00:00	0
13	2020-09-19T00:00:00	0
14	2020-09-19T00:00:00	0
15	2020-09-19T00:00:00	1
16	2020-09-19T00:00:00	0
17	2020-09-19T00:00:00	0
18	2020-09-19T00:00:00	0
19	2020-09-19T00:00:00	0

1000 rows (truncated) Prev 1 / 50 Next

EXPLANATION

The model predicts that item_id 3504 will receive around 6 views in the next 90 days, starting from 2020-09-19.

Looking at global trends and the local data around this item explains this prediction:

1. **Item characteristics:**
   - Category: "Garment Lower body" and "Trousers" within the "Divided Collection"
   - Color: Blue, Light perceived color value, Solid graphical appearance
   - None of these item attributes have direct attribution scores (all zero), indicating the model did not find these static item features critical for predicting views here.
2. **Orders linked to the item (hop 1):**
   - This item had many past orders with prices mostly around 42.38, except one at 17.35.
   - The sales_channel_id is mostly 2 for these orders.
   - In global cohorts for orders.price (hop=1), items priced around 41.86-50.73 have moderate average views target (~1.289) but lower than some lower price cohorts. However, the predicted views here (6) exceed typical cohort averages, so pricing alone doesn't explain the jump.
3. **Orders count cohorts:**
   - There are many orders linked to this item (32 orders). Globally, higher order counts at hop=1 associate with moderate view counts (mostly below 3.4 in targets).
   - The subgraph shows no orders have price or date cell scores, suggesting order details don't strongly influence the view prediction here, possibly due to high volume diluting signal.
4. **Returns and reasons:**
   - Several returns linked, mainly for "color not as expected" or "quality not as expected," but return date and reasons have zero importance in the local subgraph.
   - In Cohorts, returns counts do correlate modestly with views but aren't decisive here.
5. **Views history:**
   - Views count cohorts at hop=1 show a cohort with (4-6] views having an average target of ~1.66. This item's predicted value of 6 views is slightly above this typical range, indicating slightly higher attention.
   - The model likely integrates time proximity: views clustered around recent dates (Aug-Sep 2020) may increase predicted views.
6. **User-level features:**
   - Users linked to orders and views show variation in age and activity, mostly inactive with regular or no fashion news frequency.
   - None of these user attributes have local scores, suggesting user demographics don't strongly impact this prediction locally.
7. **Local view attribution scores:**
   - All cells in the subgraph explanation for this prediction have attribution scores of zero, indicating the prediction is driven more by broader contextual patterns learned at cohort level rather than specific attribute values of this item or its linked entities.

Summary:
The model predicts 6 views for this item based on the context of many past orders, consistent order prices, and overall typical view ranges from similar items in the dataset. While specific attributes of the item, orders, returns, or users don't strongly influence this individual prediction, the accumulated data about order counts and general temporal patterns hint at a moderately popular item with steady interest likely leading to this volume of views. This predicted level notably exceeds average views for many cohorts in the data but aligns with items in the orders count cohorts with moderate activity.

MODEL PERFORMANCE

1.959	28.749	5.362
MAE	MSE	RMSE

PQL Clear Explain Evaluate Run

```
PREDICT COUNT(returns.*, 0, 365, days)
FOR EACH users.user_id
```

5000 rows ×

ENTITY	ANCHOR_TIMESTAMP	TARGET_PRED
0	2020-09-19T00:00:00	4
1	2020-09-19T00:00:00	0
2	2020-09-19T00:00:00	0
3	2020-09-19T00:00:00	0
4	2020-09-19T00:00:00	14
5	2020-09-19T00:00:00	4
6	2020-09-19T00:00:00	0
7	2020-09-19T00:00:00	1
8	2020-09-19T00:00:00	0
9	2020-09-19T00:00:00	0
10	2020-09-19T00:00:00	0
11	2020-09-19T00:00:00	2
12	2020-09-19T00:00:00	3
13	2020-09-19T00:00:00	0
14	2020-09-19T00:00:00	2
15	2020-09-19T00:00:00	0
16	2020-09-19T00:00:00	14
17	2020-09-19T00:00:00	0
18	2020-09-19T00:00:00	9
19	2020-09-19T00:00:00	15

1000 rows (truncated) Prev 1 / 50 Next

EXPLANATION

The model predicts a notably high number of returns—about 32—for user ID 967 over the next year. Here's why the model reached this conclusion based on both global patterns and specific user data:

1. **Global Patterns from Cohorts:**
   - Users with high order counts strongly correlate with higher return counts. For instance, users who placed more than 18 orders tend to have around 8.4 returns on average, and at the second hop level, users with any orders at all show even higher return rates (target of 12.0 in the cohort of more than 0 orders).
   - Recent order dates (within 1-2 months) correspond to higher returns (globally about 6-7 returns on average), although this user's orders are mostly older, still some recent order dates have moderate target impact.
   - Certain return reasons like "quality not as expected," "color not as expected," and "too small" have similarly elevated return counts (~4 to 4.16 on average) globally.
   - Product-level features such as item sections related to "Womens Everyday Collection," and "Womens Tailoring" have moderately high average return rates (3-4 returns), with some product types like "Underwear" and "Nightwear" showing returns exceeding 4, pointing to susceptibility in those categories.
   - Users flagged with "active" and "fashion_news_frequency: Regularly" also show slightly elevated returns on average (targets about 3.49 and 3.45 respectively).
2. **Local View of This User's Data:**
   - This user is "ACTIVE" club member, "active: True," and receives fashion news "Regularly," which aligns with global cohorts that show moderately elevated return risk.
   - The user has 32 linked orders, which places them in the highest order count cohort associated with the highest return counts (target ~8.4 returns or more). The high number of orders logically leads to a higher absolute return count.
   - Orders span various sales channels but lean more heavily on sales channel 2, linked to slightly higher average return targets (11.97-12.17 at the second hop level).
   - Many orders have prices in mid to high ranges (e.g., around $25-$34), falling into cohorts with moderately high returns, especially in the mid-price categories ($26-$28 shows about 5.2 returns on average).
   - The subgraph shows numerous returns explicitly linked to this user's orders with return reasons frequently related to "quality not as expected," "color not as expected," "too small," and "changed my mind." These reasons are known global drivers of returns.
   - Items linked to returns include categories with higher return tendencies such as lingerie, swimwear, and casual women's apparel. Colors like "Black," "Dusty Light," and "Grey," noted in the subgraph, appear in cohorts with return targets around 3.2-4.6.
   - The subgraph's cell attribution scores for return dates and return reasons are moderate, indicating that individual return records contribute meaningfully to the predicted count.
3. **Summary:**
   - The prediction is largely driven by the user's high purchase activity (32 linked orders), which globally associates with higher return volumes.
   - The user's order and return history reflect common high-risk return reasons and product categories.
   - The user's engagement status (active club member, regular fashion news reader) correlates with somewhat higher return frequency.
   - The model integrates these factors, combining broad cohort tendencies with the user's rich subgraph of orders and returns, explaining the high forecast of about 32 returns.

This means the user is a heavy buyer prone to returning items, particularly in categories and for reasons that historically drive more returns. The business might focus retention or product-fit strategies on such users to reduce return-induced costs.

MODEL PERFORMANCE

3.001	28.213	5.312
MAE	MSE	RMSE